Ikhtiar Hidayatullah

NOC Engineer — Network & Automation · Skynet

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Pamekasan, East Java

SUMMARY

NOC engineer at an FTTH ISP in East Java, working on both the network and the software that runs it. Day to day I operate MikroTik routers, GPON OLTs from three vendors and 1,000+ TR-069-managed ONUs, and handle routing, PPPoE, load balancing and incident triage. I also own the company's billing and operations platform: I design the features, review every change, and ship through CI with blocking tests and guarded deploys. I care about changes that can be measured, explained and rolled back.

EXPERIENCE

NOC Engineer — Network & Automation — Skynet (FTTH ISP)

2025 — Present · East Java, Indonesia

- Operate 10+ MikroTik routers and 4+ OLTs (ZTE, Huawei, HiOSO) serving 2,000+ PPPoE subscribers.
- Fixed broken WhatsApp calls by changing PCC to per-source balancing and tuning UDP timeouts; tuned queues (multi-queue, fq_codel) and router time sync.
- Rolled out forced DNS with failover, PPPoE syslog to the server, and a captive pay page for isolated customers.
- Own the billing and operations platform (Laravel + React): designed features, reviewed every change, and shipped through CI with blocking tests and guarded deploys.
- Turned new installs and migrations into a WhatsApp flow with OCR, automatic ONU provisioning (ZTE/Huawei) and GenieACS tagging.
- Built outage detection with cause hints from OLT, ACS and PPPoE data, run in shadow mode with an accuracy report before going live.
- Security and performance hardening on routers and servers, with written rollback steps for every change.

Computer Network Installation Training — UPT BLK Surabaya

Oct 2025 — Nov 2025 · Surabaya, East Java

- Subnetting and VLSM.
- Switch, router and access point configuration; VLANs.
- Routing: static, RIP, OSPF, EIGRP, BGP.
- LAN and Wi-Fi troubleshooting.

Tech Team Academy (Internship) — PT. ARAI Rubber Seal Indonesia

Feb 2024 — Jun 2024 · Gresik, East Java

- Assembled and replaced PC units; set up and fixed network printers.
- Installed and configured CCTV.
- LAN cabling and crimping.
- Backend work on the in-house ERP (MySQL + C#).

SELECTED PROJECTS

Skynet eBilling — Billing & operations platform (Laravel + React): invoices, online payments, credit, automatic isolate/reconnect on MikroTik, isolation portal; shipped through CI with guarded deploys.

New Installs & Customer Migration over WhatsApp — WhatsApp install & migration flow: OCR of ID card/ONU label, automatic ONU provisioning on ZTE/Huawei OLTs, GenieACS tagging, auto reconciliation.

Outage Detection & Root-Cause Hints — Outage detection with cause hints from OLT (LOS/dying gasp), GenieACS Wi-Fi data and PPPoE syslog; shadow mode with an accuracy report before going live.

Network Monitoring & Report Automation — Monitoring: Playwright PNG reports to WhatsApp/Telegram, then Prometheus/Grafana with an in-app Network tab, OLT optical levels and a CPU-steal-aware job gate.

ISP PPPoE Platform on MikroTik — PPPoE baseline on MikroTik: 100/200 Mbps package profiles, multi-queue + fq_codel tuning, PPPoE syslog, forced DNS with failover.

Dual-Upstream Load Balancing & Failover (PCC + Recursive) — Dual-upstream PCC + recursive failover; fixed broken WhatsApp calls (PCC src-address, UDP timeout).

ODP/ODC Deployment Tracker — ODP/ODC deployment tracker (Vue + Express) replacing a shared spreadsheet, with charts and Excel export.

SKILLS

MikroTik / RouterOS	PPPoE server and profiles, simple queues, fq_codel, PCC load balancing, recursive failover, OSPF, VLAN, OpenVPN/L2TP tunnels, firewall/NAT, netwatch, syslog, RouterOS API
GPON / OLT	ZTE C300, Huawei MA5608T, HiOSO; ONU provisioning over CLI, optical levels, LOS / dying gasp, service ports, migrations between OLTs
TR-069 / ACS	GenieACS (NBI, provisions, virtual parameters, presets), DHCP option 43, connection requests over tunnels, FiberHome CPEs
Linux / servers	Ubuntu, Proxmox, Docker, Docker Compose, Coolify, Traefik, systemd, MySQL, Redis
Monitoring	Prometheus, Grafana, blackbox and MikroTik exporters, alert digests
Web development	Laravel, PHP, Inertia.js, React, TypeScript, Tailwind CSS, Vue 3, Node.js / Express, Astro
CI/CD & testing	GitHub Actions (self-hosted runners), PHPUnit test groups, Playwright e2e, pre-push checks, scripted deploys with guards and rollback, secret scanning
Automation	WhatsApp gateway (Evolution API), group-scoped bots, OCR (PaddleOCR, Tesseract), Python + Playwright reports, Midtrans / Duitku payment callbacks
Ways of working	Written plans before changes, dry-run → backup → apply → revert for data fixes, runbooks, code review before every merge

EDUCATION

Universitas Trunojoyo Madura — B.Sc. Informatics Engineering — GPA 3.70 / 4.00 `Jul 2020 – Oct 2025`

CERTIFICATES

Pemasangan Jaringan Komputer — UPT BLK Surabaya — UPT Balai Latihan Kerja Surabaya · Okt 2025 · ID 25005.09.035.02.2025

Network Penetration Testing — Cyber Academy Indonesia · Agu 2024 · ID NTPT0109243207

Fundamentals of DevOps — Dicoding Indonesia · Sep 2023